

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Miniaturized GC/MS Instrumentation: MEMS-based Gas Chromatography Coupled with Miniature Quadrupole Array and Paul Ion Trap Mass Spectrometers

Authors: Paul M. Holland, Ara Chutjian, Murray Darrach, Otto Orient

Institute: Jet Propulsion Laboratory

ABSTRACT

Miniaturized chemical instrumentation is needed for in situ measurements in planetary exploration, spaceflight applications, and other extreme environments. Here, factors such as small size and enhanced robustness are important. In response to this need, we are developing miniaturized GC/MS instrumentation which combines chemical separations by gas chromatography (GC) with mass spectrometry (MS) to provide positive identification of chemical compounds in complex mixtures of gases, such as those found in the International Space Station's cabin atmosphere. Our designs utilize MEMS-based micro gas chromatography components coupled with either a miniature quadrupole mass spectrometer array (QMSA) or compact, high-resolution Paul ion trap. Key design issues include high sensitivity, good MS resolution (0.5 amu FWHM or better), low power, robustness, low GC flow rates to minimize vacuum-pumping requirements, and the use of a modular approach to adapt to different environments. Among the potential applications for such instrumentation are in situ detection of astrobiology signatures (using air sampling or ground-drilling techniques), planetary aeronomy, and monitoring of cabin air during long duration human flight. This work has been carried out in part at the Jet Propulsion Laboratory, California Institute of Technology, and was supported by the National Aeronautics and Space Administration.